Session 5: Joins and RelationshipsTrainer Coverage:Foreign keys and primary keys.Inner Join, Left Join, Right Join.Demo: Link users with their transactions table.Pending

Tasks

1) **Create two tables in SQL: influencers (with influencer_id as PRIMARY KEY and name) and posts (with post_id as PRIMARY KEY, influencer_id as FOREIGN KEY, and caption). Insert at least 3 influencers and 2 posts for each influencer.**

   Query:

```sql
CREATE TABLE influencers (
    influencer_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL
);

CREATE TABLE posts (
    post_id INT PRIMARY KEY AUTO_INCREMENT,
    influencer_id INT,
    caption VARCHAR(255),
    FOREIGN KEY (influencer_id) REFERENCES influencers(influen
);

INSERT INTO influencers (name) VALUES
('abhi'),
('velcy'),
```

```sql
('abhi'),
('velcy'),
('harish');

INSERT INTO posts (influencer_id, caption) VALUES
(1, 'My first travel vlog!'),
(1, 'Exploring a new place today!'),

(2, 'Morning workout routine!'),
(2, 'Healthy food ideas!'),

(3, 'New technology tips!'),
(3, 'Learning SQL today!');
```

Run SQL query/queries on database my database:

```sql
SELECT * FROM influencers;

SELECT * FROM posts;
```

n SQL query/queries on table my database.posts:

```sql
SELECT
    influencers.name,
    posts.caption
FROM influencers
JOIN posts
ON influencers.influencer_id = posts.influencer_id;
```

Output:

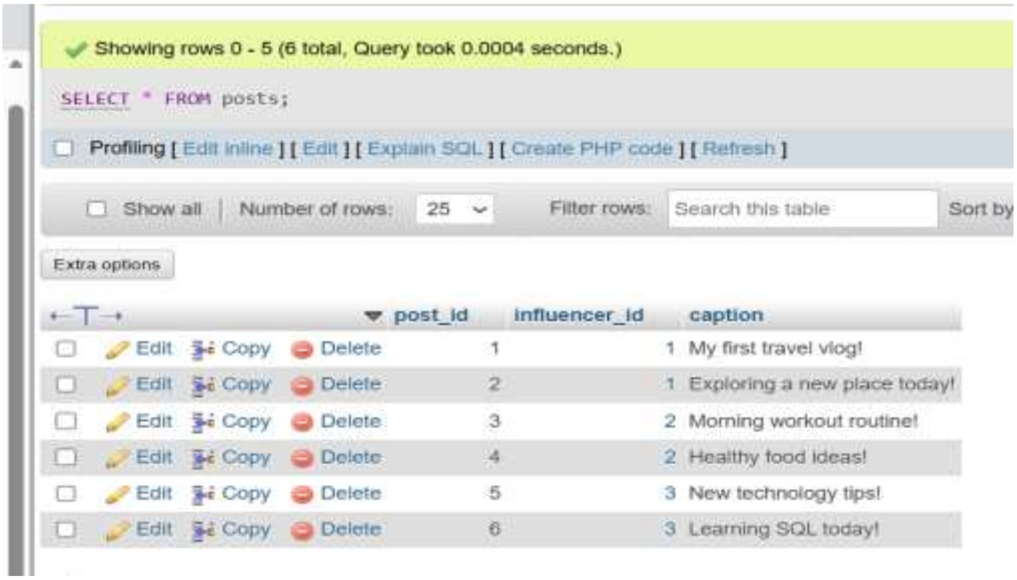
```sql
SELECT * FROM influencers;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refre

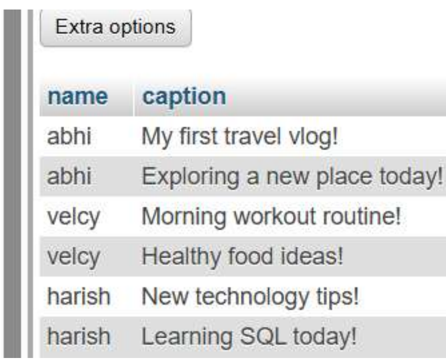
Show all | Number of rows: 25 Filter rows: Search

Extra options

	influencer_id	name
Edit Copy Delete	1	abhi
Edit Copy Delete	2	velcy
Edit Copy Delete	3	harish



	post_id	influencer_id	caption
Edit Copy Delete	1	1	My first travel vlog!
Edit Copy Delete	2	1	Exploring a new place today!
Edit Copy Delete	3	2	Morning workout routine!
Edit Copy Delete	4	2	Healthy food ideas!
Edit Copy Delete	5	3	New technology tips!
Edit Copy Delete	6	3	Learning SQL today!



name	caption
abhi	My first travel vlog!
abhi	Exploring a new place today!
velcy	Morning workout routine!
velcy	Healthy food ideas!
harish	New technology tips!
harish	Learning SQL today!

2) **Write an SQL query using INNER JOIN to display each post's caption along with the name of the influencer who posted it, based on the tables you created.**

Query:

```sql
SELECT
    posts.caption,
    influencers.name
FROM posts
INNER JOIN influencers
ON posts.influencer_id = influencers.influencer_id;
```

Output:

Show all | Number of rows: 25

Extra options

caption	name
My first travel vlog!	abhi
Exploring a new place today!	abhi
Morning workout routine!	velcy
Healthy food ideas!	velcy
New technology tips!	harish
Learning SQL today!	harish

3) **Write an SQL query using LEFT JOIN to show all influencers and their posts, including influencers who haven't posted anything yet. If an influencer has no posts, display 'No Posts' in the caption column.

Hint: Use IFNULL or COALESCE to handle NULL values in the result.**

Query:

```sql
SELECT
    influencers.name,
    COALESCE(posts.caption, 'No Posts') AS caption
FROM influencers
LEFT JOIN posts
ON influencers.influencer_id = posts.influencer_id;
```

Output:

```
SELECT influencers.name, COALESCE(posts.caption, 'No P
posts.influencer_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code]

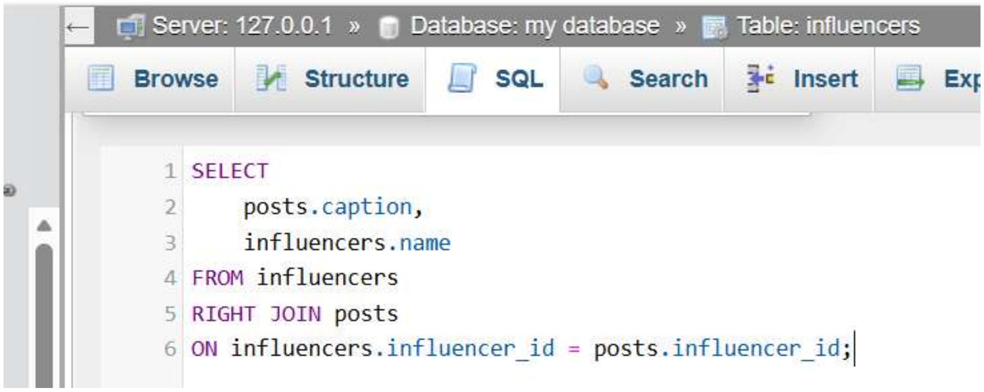
Show all | Number of rows: 25 Filter rows:

Extra options

name	caption
abhi	My first travel vlog!
abhi	Exploring a new place today!
velcy	Morning workout routine!
velcy	Healthy food ideas!
harish	New technology tips!
harish	Learning SQL today!

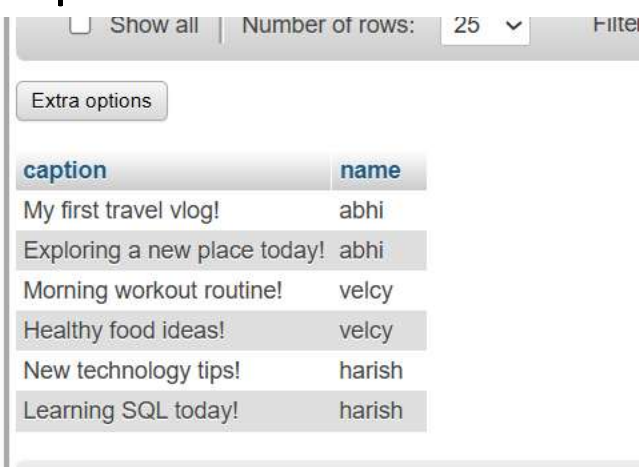
4) **Write an SQL query using RIGHT JOIN to list all posts and the corresponding influencer's name, ensuring that even posts without a matching influencer_id (if any) are shown.**

Query:




```sql
SELECT
    posts.caption,
    influencers.name
FROM influencers
RIGHT JOIN posts
ON influencers.influencer_id = posts.influencer_id;
```

Output:



caption	name
My first travel vlog!	abhi
Exploring a new place today!	abhi
Morning workout routine!	velcy
Healthy food ideas!	velcy
New technology tips!	harish
Learning SQL today!	harish